School of Mechanical Engineering
ME 30801 — **Fluid Mechanics Laboratory**
Fall 2025
Syllabus

Course Information

- **CRN:** 22832, 22837, 22838
- **Prerequisite(s):** ME308 – Fluid Mechanics
- **Lab Preparation Lecture Meeting Location/Information***
  - Lab 001: Merge; used for posting lab material and submitting assignments onto Brightspace
  - Lab 002: Tuesday, 1:30-2:20PM, ME 1130
  - Lab 003: Tuesday, 4:30-5:20PM, BHEE 129
- **Dates of Instruction:** August 25, 2025 – December 13, 2025
  (Finals December 15-20, 2025; semester ends December 20, 2025)
- **Observed holidays on academic calendar**
  - Labor day: September 1, 2025
  - Fall Break: October 13-14, 2025
  - Thanksgiving: November 26-29, 2025
- **Course credit Hours:** 1 Credit Hour
- **Fluid Mechanics Lab – Team Times and Expectations**
  - Lab hour reserved by each team (50 mins per week) @ Potter 160 OR ME 1032
- **References**
  - Pritchard, P.J., *Fox and McDonald's Introduction to Fluid Mechanics*, 9th ed., Wiley & Sons.
  - There are other undergraduate fluid mechanics texts available in the Library of Engineering and Science located in WALC and the internet. Students are encouraged to use these additional sources as references.
  - Undergraduate calculus and thermodynamics textbook.

* Lectures will be in person and only in weeks indicated in the progress schedule. Lecture attendance is mandatory.

Note:

1. Each lab team will get help from and work with a designated TA for all the standard labs.
2. Office hours of each TA will be announced in Brightspace.
3. Students should receive a response from the TA or instructor within 24 hours of emailing via Purdue account.
4. Any issues from interaction with TAs should be reported to and resolved by the instructor.
5. Labs 2, 3, and 4 will be in Potter 160; labs 1 and 5 will be in ME 1032.
6. **Piazza:** https://piazza.com/purdue/fall2025/me30801; We will use Piazza to answer lab related questions and as a vehicle to identify lab group members

Course Description

This course is the lab component of an already established ME 30900 Fluid Mechanics (4 cr) course, which is now being split into an ME 30800 (3 cr) Fluid Mechanics lecture course, and the material covered in this new ME 30801 (1 cr) Fluid Mechanical Laboratory. The content of both courses remains unchanged. The splitting of the lab from the lecture will enable better enrollment load balancing in the existing ME undergraduate laboratory facilities. Details of this new laboratory are outlined in the appended material below.

Learning Outcomes

1. Understand the technical background and fundamental concepts of experimental techniques in fluid mechanics.
2. Apply fluid mechanics principles to the design and analysis of flow systems and experiments.
3. Conduct physical experiments and master basic laboratory test skills.
4. Analyze experimental data using appropriate methods.
5. Communicate findings effectively through written lab reports.
6. Complete a laboratory design project to reinforce the engineering design process.

Computing resources and usage

- Access to the internet is required for this course.
  - *Brightspace*: the site for course documents, such as administrative materials, notes, quizzes, lab manuals, teaching videos, and other materials.
  - *Gradescope*: the site for submitting and returning assignments; for Gradescope guidance, please refer
    - to https://www.gradescope.com/get_started
  - *Zoom* and *Webex*: may be used for virtual office hours by the instructor or TAs.
  - *Purdue Email*: course email correspondence will be only with your @purdue.edu email address.
- Knowledge of word processing for drafting a professional report.
- Knowledge of basic programming (for example, MATLAB, and Python) and/or technical plotting software (e.g., Tecplot) will be necessary for processing the experimental data and preparing lab reports.
- ***Processing experimental data and generating plots using EXCEL is prohibited!*** Everyone should practice coding skills (MATLAB, Python, etc.) for data processing,

Course-related announcements

All course-related announcements made by the instructor and lead TA will be sent out and archived via Brightspace. The students may trace all announcements in Brightspace. The teaching team is NOT responsible for any consequences of neglected announcements.

Attendance Policy

Lecture/lab attendance is required. This course follows Purdue's academic regulations regarding attendance, which state that students are expected to be present for every meeting of the classes in which they are enrolled. Attendance will be taken at the beginning of each class and lateness will be noted. When conflicts or absences can be anticipated, such as for many University-sponsored activities and religious observations, the student should inform the instructor of the situation as far in advance as possible. For unanticipated or emergency absences when advance notification to the instructor is not possible, the student should contact the instructor as soon as possible by email or phone. When the student is unable to make direct contact with the instructor and is unable to leave word with the instructor's department because of circumstances beyond the student's control, and in cases falling under excused absence regulations, the student or the student's representative should contact or go to the Office of the Dean of Students (ODOS) website to complete appropriate forms for instructor notification. Under academic regulations, excused absences may be granted by ODOS for cases of grief/bereavement, military service, jury duty, parenting leave, or emergent or urgent care medical care.

Laboratory Safety

Lab safety is of paramount importance. The safety issues may include but are not limited to, electrical safety, proximity to fast-moving equipment, working in confined spaces, exposure to strong light, etc. In addition, all the lab equipment and instruments are very expensive to replace. All personnel working in the lab should abide by all the requirements to maintain a safe working environment. Please refer to the TA or instructor for any safety concerns.

Laboratory Teams

Every student must join a team of three voluntarily. Instructor's approval must be pursued to form a team of four (not encouraged). Members from the same team will receive the same score for the team score category. Each team should reserve the lab hours following instructions. *Each team is assigned a unique label indicating the team number, e.g., T02, T13, etc*. ***This label should appear in all submissions.***

Note: It is the responsibility of every team to check that the reserved lab hours do not conflict with their lecture section. Members from may be from different lecture sections but the lecture sessions of every team member should NOT have conflict with the scheduled lab hour.

Assignments

Students are responsible for finishing and submitting the following assignments in addition to finishing the lab work. Please refer to **Grading Policy** and **Progress Schedule** for more details.

- Prelab Quizzes
- Lab Reports
- Draft Proposal for Open Lab Project
- Revised Proposal for Open Lab Project
- Report and Presentation Video for Open Lab Project

Grading Policy

Final grades will be determined using the following algorithm.

1. All final scores may be curved at the instructor's discretion.
2. The final grades will be determined using the following percentile table.

A+: 97 ≤ score	A: 93 ≤ score < 97	A-: 90 ≤ score < 93
B+: 87 ≤ score < 90	B: 83 ≤ score < 87	B- : 80 ≤ score < 83
C+: 77 ≤ score < 80	C: 73 ≤ score < 77	C-: 70 ≤ score < 73
D+: 67 ≤ score < 70	D: 63 ≤ score < 67	D-: 60 ≤ score < 63
F: score < 60		

10% Prelab Quiz (Individual Score)

Prelab quizzes will be assigned and collected via Brightspace.
Late submissions will NOT be accepted for any reason and will get zero points. There will be two attempts offered, and the quiz score will be the highest one.
*If the quiz score is less than 70%, the student will **NOT be allowed to participate in the lab test** and get a **zero** in the lab performance for that lab.*

45% Standard Lab (Labs 1-5, Team Score)

a) Every student must join a team of three (or four with the instructor's approval). ***The same team should work together for all lab efforts for the entire semester***. All the members will receive the same scores for this part of standard labs. Students can form teams across lecture sections.
b) Every team member must finish the corresponding prelab quiz individually.
c) Each team needs to work independently in the lab following the instructions from the lab manuals, under the supervision of the TA.
d) The instructor and/or TA will observe and document the performance of every team member during each lab section.
e) Each team should work together to process the data acquired from the experiments and write the lab report. *The contribution from every team member should be indicated at the end of the report.*
f) ***The lab report should strictly follow the format requirement (font, structure, words limit, etc) indicated in the template***.
g) The lab report will be submitted and returned electronically through Gradescope. *No late lab report will be accepted for any reason*.

h) In addition to correct technical content, lab reports should be presented professionally following the guideline in the report template. Technical content includes material such as methodology, equipment/instruments, key elements for data analysis (assumptions, equations, control volumes, free body diagrams, …), figures, numerical values, and units. The communications content includes items such as the document's legibility and formatting, presentation of plots, tables, and figures, and text describing the approach. All reports must be prepared using word processing software and submitted as PDF file.
i) Templates for preparing the first few reports and rubrics for grading will be posted in Brightspace for reference.
j) Out of the five standard labs (Labs 1 to 5), ***the lowest report score will account for 5% while the other four will account for 10%, respectively, in calculating the final grade****.*

25% Open Lab (Team Score)

a) Revised Project Proposal: 5%; Lab Report and Presentation: 20%.
b) Each team needs to design an open lab project that is related to fluid mechanics.
c) A draft proposal will be reviewed and approved by the instructor (**submission deadline: middle of the semester, Sunday October 26**). Certain revision comments will be given. The instructor may request an individual meeting with a certain team to discuss the project idea.
d) Every team must submit a final (revised proposal) for grading.
e) ***The difficult level of the open lab project will be rated in the final grade***. The difficulty level will be evaluated when the draft proposal is reviewed. The grading rubrics will be posted in Brightspace for reference.
f) The finalized project will be conducted independently by the team under the supervision of the TA.
g) Each team needs to submit a project report and a recorded presentation to introduce the project. Details will be announced.

8% Lab Performance (Individual Score)

The TA will observe and record everyone's performance in each lab and give individual scores on attendance, punctuality, and professional manner (handling lab equipment correctly, showing respect to others, following safety rules, cleaning up after lab, etc.). Grading rubrics will be posted in Brightspace.

7% Peer Evaluation (Individual Score)

Everyone in a lab team needs to contribute to all the lab efforts in an approximately equal way. There will be 2 peer evaluation surveys (completed by the teammates) after lab 2 and lab 5.

5% Instructor's Discretion (Individual Score)

a) 2% is given to anyone who completes an anonymous exit survey and submits the proof of submission near the end of the semester.
b) 3% is assigned based on metric including lecture attendance record and syllabus quizzes, as well as demonstrated leadership in teamwork.
c) Instructor discretion points are not transferrable between sections.

Notes

1. **Assignment re-grade requests** must be submitted within <u>one week (seven days)</u> from the date the graded document has been made available for return. Re-grades submitted after this deadline will NOT be considered. Re-grade requests must include a statement detailing the justification for the re-grade. Note that the item to be re-graded is re-graded from scratch and may result in a score lower than the original score.
2. ***Each team must finish the labs and reports independently. Sharing data and/or copying reports of other teams is strictly prohibited***.
3. ***All submitted reports will go through a plagiarism checking. Identical materials (writings without appropriate citation, plots, etc.) with the ones in reports from previous semesters will be counted as plagiarism.***

4. ***Violation of notes 2 and 3 will result in a score of zero for the entire portion of the corresponding lab*** and all individuals involved will be reported to the Office of the Dean of Students for academic dishonesty.
5. For privacy, scores will NOT be reported via e-mail or telephone. Scores will be disseminated via *Brightspace* and *Gradescope* only.
6. **Copyright**: Online educational environments, like all learning environments, should provide opportunities for students to reflect, explore new ideas, post opinions openly, and have the freedom to change those opinions over time. Students enrolled in and instructors working in online courses are the authors of the works they create in the learning environment. As authors, they own the copyright in their works subject only to the university's right to use those works for educational purposes (Visit Purdue University Copyright Office). Students may not copy, reproduce or post to any other outlet (e.g., YouTube, Facebook, or other open media sources or websites) any work in which they are not the sole or joint author or have not obtained the permission of the author(s).
    - Course materials and commercial web pages: In general, class notes (including answer keys to homework) are "derivative works" of the instructor's presentations and materials and are subject to the instructor's copyright. As such, they cannot be sold or bartered (for example, on commercial web pages) without express written permission from the instructor.
7. **The Protect Purdue Plan**, which includes the Protect Purdue Pledge, is campus policy and as such all members of the Purdue community must comply with the required health and safety guidelines. If non-compliance continues, possible results include instructors asking the student to leave class and instructors dismissing the whole class. Students who do not comply with the required health behaviors are violating the University Code of Conduct and will be reported to the Office of the Dean of Students with sanctions ranging from educational requirements to dismissal from the University.
8. **Classroom Guidance Regarding Protect Purdue**: Any student who has substantial reason to believe that another person in a campus room (e.g., classroom) is threatening the safety of others by not complying (e.g., not wearing a mask) may leave the room without consequence. The student is encouraged to report the behavior and to discuss the next steps with their instructor. Students also have the option of reporting the behavior to the Office of the Student Rights and Responsibilities. See also Purdue University Bill of Student Rights.

Classroom Guidance regarding Protect Purdue

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Any student who has substantial reason to believe that another person is threatening the safety of others by not complying with Protect Purdue protocols is encouraged to report the behavior to and discuss the next steps with their instructor. Students also have the option of reporting the behavior to the Office of the Student Rights and Responsibilities. See also Purdue University Bill of Student Rights and the Violent Behavior Policy under University Resources in Brightspace.

Academic Integrity and Professional Responsibility

The Purdue Honor Pledge: "*As a Boilermaker pursuing academic excellence, I pledge to be honest and true in all that I do. Accountable together - we are Purdue.*"

Academic integrity is one of the highest values that Purdue University holds. Individuals are encouraged to alert university officials to potential breaches of this value by either emailing integrity@purdue.edu or by calling 765-494-8778. While the information may be submitted anonymously, the more information is submitted the greater the opportunity for the university to investigate the concern. More details are available on our course Brightspace under University Policies.

Incidents of academic misconduct in this course will be addressed by the course instructor and referred to the Office of Student Rights and Responsibilities (OSRR) for review at the university level. Any violation of course policies as it relates to academic integrity will result minimally in a failing or zero grade for that particular assignment, and at the instructor's discretion may result in a failing grade for the course. In addition, all incidents of academic misconduct will be forwarded to OSRR, where university penalties, including removal from the university, may be considered.

Nondiscrimination Statement

Purdue University is committed to maintaining a community that recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life. A hyperlink to Purdue's full Nondiscrimination Policy Statement is included in our course Brightspace under University Policies.

Mental Health/Wellness Statement

If you find yourself beginning to feel some stress, anxiety, and/or feeling slightly overwhelmed, try WellTrack**.** Sign in and find information and tools at your fingertips, available to you at any time.

If you need support and information about options and resources, please contact or see the Office of the Dean of Students. Call 765-494-1747. Hours of operation are M-F, 8 am- 5 pm.

If you find yourself struggling to find a healthy balance between academics, social life, stress, etc., sign up for free one-on-one virtual or in-person sessions with a Purdue Wellness Coach at RecWell. Student coaches can help you navigate through barriers and challenges toward your goals throughout the semester. Sign-up is free and can be done on BoilerConnect.

If you're struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 during and after hours, on weekends and holidays, or by going to the CAPS office on the second floor of the Purdue University Student Health Center (PUSH) during business hours. The CAPS website also offers resources specific to situations such as COVID-19.

Basic Needs Security

Any student who faces challenges securing their food or housing and believes this may affect their performance in the course is urged to contact the Dean of Students for support. There is no appointment needed and Student Support Services is available to serve students 8 a.m.-5 p.m. Monday through Friday. Considering the significant disruptions caused by the current global crisis as it relates to COVID-19, students may submit requests for emergency assistance from the Critical Need Fund.

Accessibility

Purdue University strives to make learning experiences as accessible as possible. If you anticipate or experience physical or academic barriers based on disability, you are welcome to let me know so that we can discuss options. You are also encouraged to contact the Disability Resource Center at: drc@purdue.edu or by phone: at 765-494-1247.
Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 and http://www.purdue.edu/caps (during and after hours, on weekends and holidays), or through counselors located in PUSH (business hours).

Emergency Preparedness

In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control.

Relevant changes to this course will be posted onto the course website or can be obtained by contacting the instructors or TAs via email or phone. You are expected to read your @purdue.edu email on a frequent basis.